MZUZU UNIVERSITY

FACULTY OF SCIENCE, TECHNOLOGY AND INNOVATION

DEPARTMENT OF INFORMATION AND COMMUNICATION TECHNOLOGY

BACHELOR OF SCIENCE IN INFORMATION AND COMMUNICATION TECHNOLOGY

BICT 1202: PROGRAMMING IN C

END OF SEMESTER EXAMINATION

DATE: MARCH 2021 **TIME ALLOWED: 3 HOURS**

INSTRUCTIONS

i. There are **15** questions on this paper. **ANSWER ALL.**

i. There are **TWO** pages. **PLEASE CHECK**

iii. Marks are indicated for each question.

1. Define the following terms:
   a. Function. **[1 Mark]**
   b. Object code. **[1 Mark]**
   c. Recursion. **[1 Mark]**
   d. Parameter. **[1 Mark]**
   e. Compiler. **[1 Mark]**
   f. Array. **[1 Mark]**

2. Given numbers from 1 to 30. Write a program in C that determines whether the number is even or odd. Implement using the following;
   a. **for** loop **[3 Marks]**
   b. **while** loop **[3 Marks]**
   c. **do while** loop **[3 Marks]**

3. What is the difference between *call by value* and *call by reference*? **[4 Marks]**

4. Describe how the program works and determine the output of the following program:

```
#include<stdio.h>
main(void)
{
        int count =1;
        while (count<=10)
        {
                printf("%s\n", count%2 ? "++++" : "*****");
                ++count;
                }
return 0;
}
```

[7 Marks]

5. Differentiate source code from object code. **[2 Marks]**

6. How do you declare a variable that holds a string values? **[3 Marks]**

7. Write a program in C to calculate and print the square and cube of numbers from 1 to 20. Implement square and cube as programmer defined functions. **[6 Marks]**

8. Write a program to implement an array, initialize it with values through the keyboard and search for a particular element sequentially. Make an assumption of what the element to be searched will be. **[8 Marks]**

9. What is the difference between array element 8 and the eighth element of an array? **[4 Marks]**

10. Using a single-subscripted array, write a C program that reads in *20* integer numbers, each of which is between *10* and *100*, inclusive. As each number is read, print it only if it is *not* a duplicate of a number already read. **[7 Marks]**

11. What is the significance of an algorithm to C programming? **[2 Marks]**

12. What is the advantage of an array over individual variables? **[2 Marks]**

13. Write a function ***qualityPoints*** that inputs a student's average and returns *4* if a student's average is between *90* and *100*, *3* if the average is between *80* and *89*, *2* if the average is between *70* and *79*, *1* if the average is between *60* and *69*, and *0* if the average is lower than *60*. **[8 Marks]**

14. Write a program in C that accepts grades for a course that 12 students wrote. The program should compute and print mean and percentage grade for each course. **[7 Marks]**

15. Write a program in C that solves a quadratic equation if the user inputs **a**, **b** and **c**. **[6 Marks]**

END OF QUESTION PAPER